

Symmetric · plausibly quantum-resistant

Collision-resistant hash

no quantum algorithm is known
that breaks it

STARKs

(this work)

Number-theoretic · broken by Shor

Elliptic-curve DLP

e.g. Bulletproofs

Pairings / KoE /

unknown-order groups
e.g. Groth16, PLONK

Shor's algorithm dismantles these

(factoring & discrete log fall in poly time)

The assumption family alone decides whether a system is transparent and whether it survives a quantum adversary.